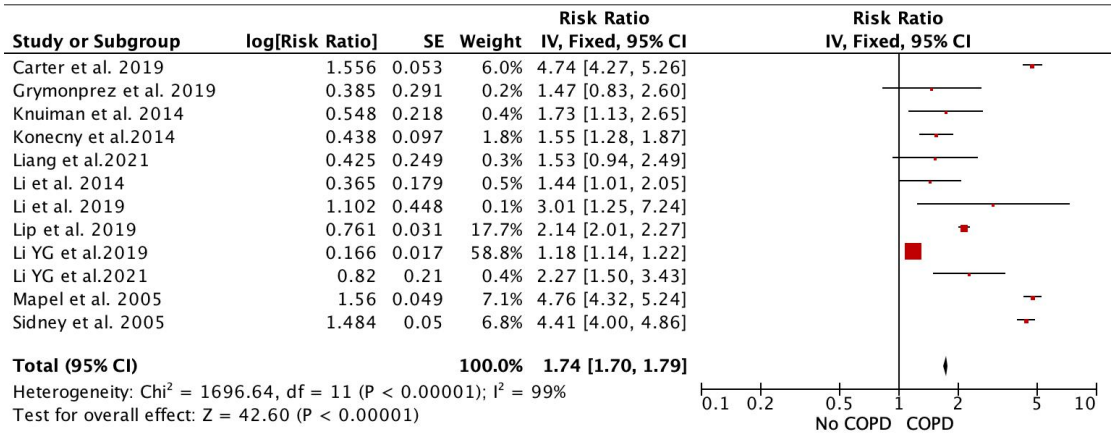
Supplementary Table 1. Retrieval strategy in the Pubmed database

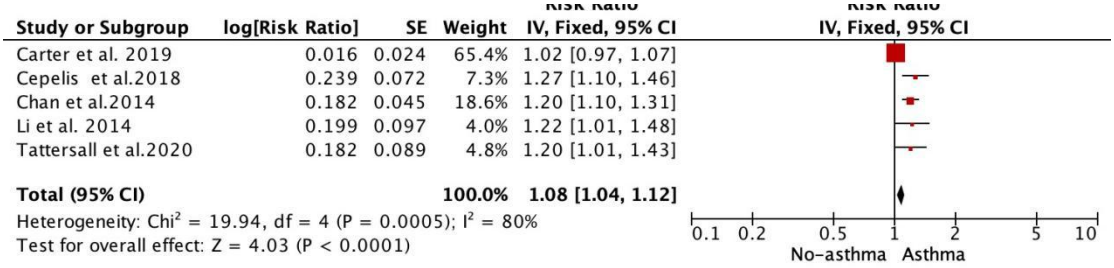
Search number	Query	Results
1	COPD	97,359
2	chronic obstructive pulmonary disease	89,748
3	asthma	203,581
4	lung function	340,944
5	pulmonary disease	1,233,863
6	1 or 2 or 3 or 4 or 5	1,485,572
7	atrial fibrillation	95,210
8	6 and 7	6,122

Supplementary Table 2. Quality assessment of the included studies based on the NOS tool

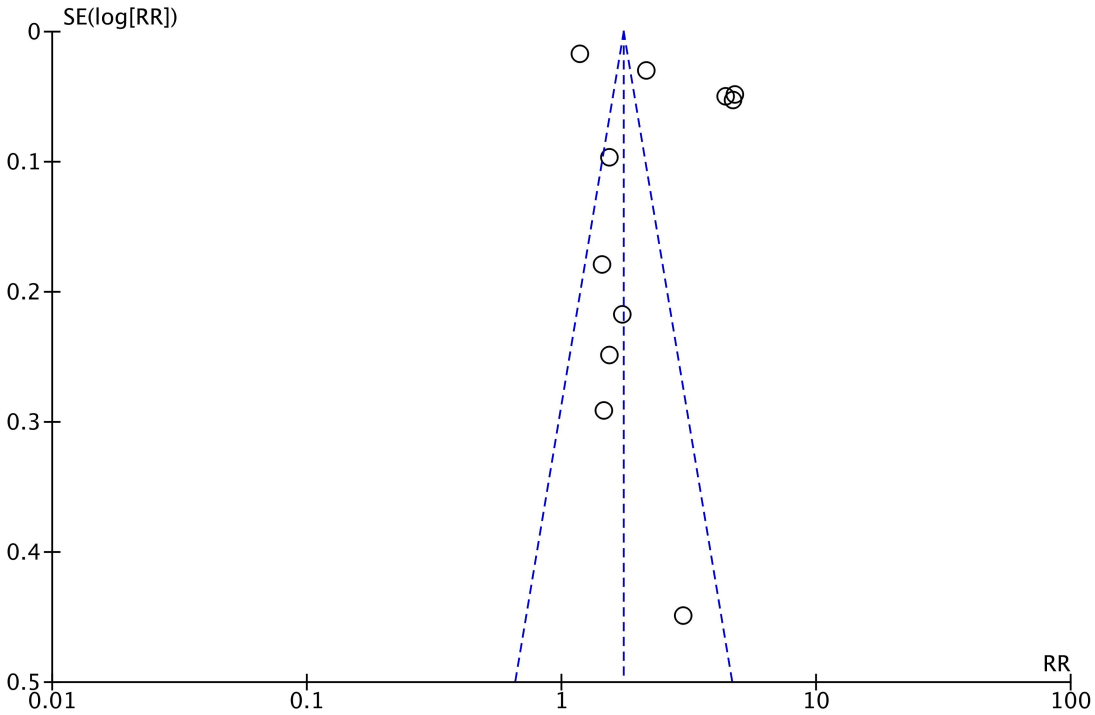
Studies	Selection				Comparability	Outcome			Total
	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest		Assessment of outcome	Length of follow-up	Adequacy of follow up	
Knuiman et al. 2014	*	*	*		**	*	*	*	8
Lip et al. 2019	*	*	*		**	*	*	*	8
Li et al. 2019	*	*	*		**	*	*	*	8
Grymonprez et al. 2019	*	*	*		**	*		*	7
Sidney et al. 2005	*	*	*		**	*		*	7
Carter et al. 2019	*	*	*		**	*	*	*	8
Konecny et al.2014	*	*	*		**	*	*	*	8
Mapel et al. 2005	*	*	*		**	*	*	*	8
Li YG et al.2019	*	*	*		**	*		*	7
Tattersall et al.2020	*	*	*		**	*		*	7
Chan et al.2014	*	*	*		**	*	*	*	8
Cepelis et al.2018	*	*	*		**	*		*	7
Li YG et al.2021	*	*	*		**	*	*	*	8



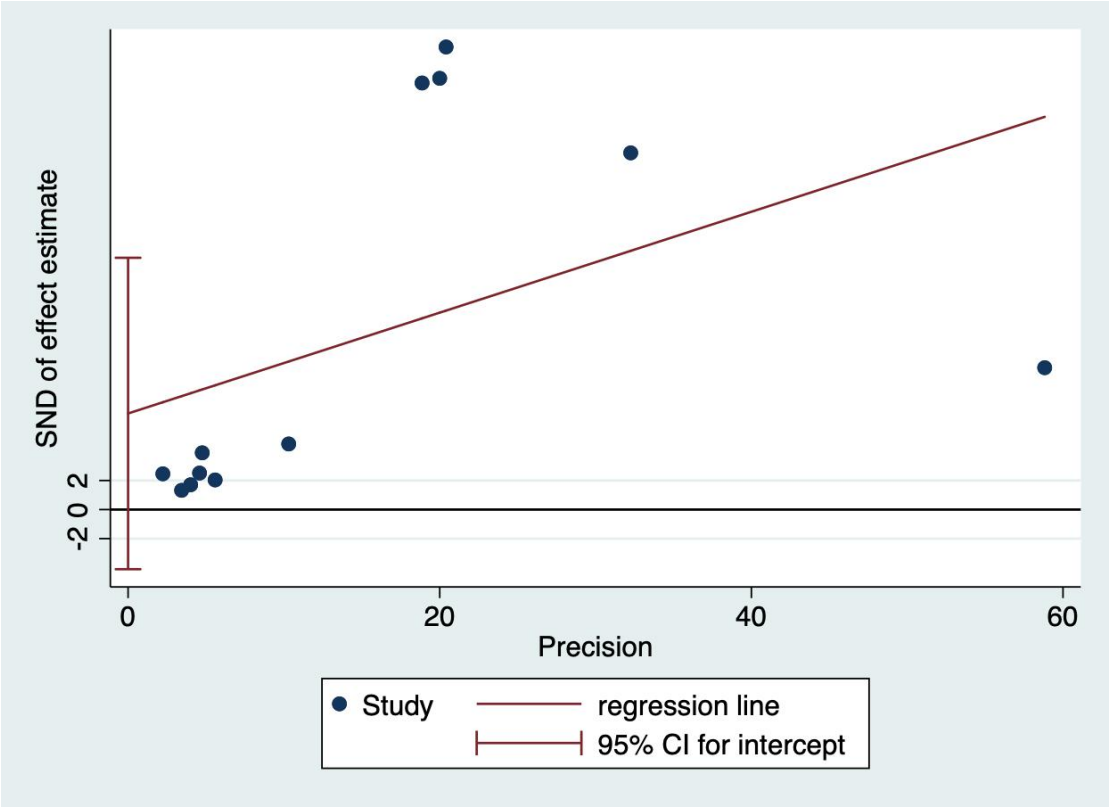
Supplementary Figure 1. Forest plot for the relationship between chronic obstructive pulmonary disease and the development of atrial fibrillation (fixed-effects model)



Supplementary Figure 2. Forest plot for the relationship between asthma and the development of atrial fibrillation (fixed-effects model)



Supplementary Figure 3. Funnel plot for chronic obstructive pulmonary disease and the development of atrial fibrillation



Supplementary Figure 4. The Egger's test for chronic obstructive pulmonary disease and the development of atrial fibrillation (P=0.199)